

Teo Yu Qi

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WORK

Software Engineer

Feb 2026 – Present

PayPal (Core Platforms, Global Wallet Solutions)

Singapore

- Built the client and backend service for merchant Amazon-PayPal Virtual Bank Account linking via Rutter, including a concurrency fix that prevents duplicate active connections from multi-tab authorization races.
- Engineered a hybrid-search RAG knowledge base (Milvus vector store, served via an MCP server) for team documentations, giving AI coding assistants single-query retrieval across previously siloed sources.
- Parallelized the orchestration service's end-to-end test suite and resolved thread-safety issues exposed by concurrent execution, cutting CI runtime from ~12 to ~5 min and eliminating flaky tests.
- Instrumented OpenTelemetry tracing and structured logging with merchant and request context across the orchestration service's backend APIs, improving observability and cutting incident triage time.

Software Engineer

Jun 2024 – Jan 2026

Shopee (Engineering Infrastructure)

Singapore

- Developed and maintained Shopee's infrastructure TechOps platform, serving over 3,000 employees.
- Engineered backend middleware microservice to serve different versions of frontend assets, enabling controlled feature rollouts for grayscale release, and A/B testing.
- Optimized frontend performance by injecting Redis-cached runtime configs into HTML templates, eliminating 5 JSON config file requests and improving average framework load time by 25.7%.
- Improved micro-frontend performance by preloading commonly-used bundles, reducing average framework load time by 5.8%.
- Alleviated engineer pain point by creating a serverless FaaS function to easily toggle release configurations via chat commands on Shopee's internal messaging platform.

Software Engineer Intern

Jan 2023 - Nov 2023

TikTok (Global E-Commerce, Data Intelligence)

Singapore

- Developed and maintained TikTok Shop frontend systems serving over 500,000 weekly active users.
- Revamped Creator Dashboard UI on mobile, contributing to +33.54% user views and +57.66% page views.
- Instrumented TikTok Shop frontend API monitoring and alerting via a centralized Axios interceptor reporting to the in-house logging framework, improving observability and reliability.
- Introduced asynchronous data export feature on Partner Compass SEA, with 70 TikTok Shop Partners utilizing the feature and ~650 file exports in the first week of release.

OTHERS

Open-Source Contributor, Electron

Nov 2025

- Improved BaseWindow API documentation by clarifying previously undocumented behavior.
- Changes merged into Electron versions 39-x-y, 40-x-y. [PR #48931]

Student Volunteer, Centre for Computing for Social Good & Philanthropy

May 2022 – Jun 2022

- Set weekly quizzes for OTH4199 AI Primer Course. (Beneficiary: Autism Resource Centre - Singapore).

PROGRAMMING LANGUAGES & FRAMEWORKS

Fluent in: TypeScript, JavaScript, Python, Java, React, Node.js, NestJS, Spring Boot, Express, HTML, CSS

Also used: Bash, C, C++, Electron, Go, MySQL, NextJS, PostgreSQL, Redis

EDUCATION

National University of Singapore (NUS)

Aug 2020 – May 2024

Bachelor of Computing in Computer Science (with Honors)

Singapore